

# Lab 3

(Due: May 27)

**Remind:** in all the lab questions, please include the screen shots as evidence of your solutions.

## 0. Submit the coding solution for Q4 in Lab Assignment 2.

1. In this problem, you will get familiar with ip format. Start the Wireshark and run

**ping www.utoronto.ca**

and then stop Wireshark. Ping **www.utoronto.ca** is to send an icmp packet. Check the first echo request packet in the Wireshark window and answer the following questions.

- a. Look at the ip header, what is the source and destination ip address?
- b. What is the upper layer protocol in ip header?
- c. what is the ip header length?
- d. Calculate the payload length for ip packet. This is **totallength - headerlength**.
- e. what is the TTL value and what is its meaning?
- f. find out which field shows the ip header is in ipv4 or ipv6 format.

2. Start Wireshark on your VM. Next, run command **sudo dhclient -r -v** and then **sudo dhclient** and finally stop Wireshark. Command **sudo dhclient -r -v** will release your current ip address. Then **sudo dhclient** will execute the DHCP protocol. Use packets in Wireshark from executing DHCP to answer the following questions.

- a. Confirm that the transport layer protocol of DHCP protocol is UDP. To do this, check a packet with DHCP protocol data and look at the transport layer header. Think about why it is not TCP (recall that TCP needs to establish a connection before exchanging messages).
- b. In addition to offer the ip address to your computer, DHCP can in fact provide you more useful configuration. Check DHCP **offer packet** to find out the following information.

**DHCP server IP:** you need this to extend your time to use the current IP address.

**Subnet mask:** this tells you the subnet type.

**Router IP:** That is the ip address your outgoing packet will first go to.

**DNS IP:** this is the ip address of the DNS server that you will request to resolve your DNS query. That is, this is your **local** DNS server.

3. In this exercise, you will look in the arp protocol execution. First, run **arp** to find out the list of records in the arp table. Next, start your wireshark and run **sudo arp -d routerIP** to delete the record of *routerIP*. Here routerIP is the **Router IP** obtained in the previous DHCP experiment. Then, you should see your VM is now starting to run arp.

a. Find our arp broadcast from your VM. What is the upper layer protocol in the link layer header? What is the broadcast MAC address? What is the ip address for which your broadcast message is intended to find out the MAC address?

b. look at the response packet for the ARP query. What is the ip address of the sender? What is its MAC address?

4. Run wireshark and access [www.example.com](http://www.example.com) and stop Wireshark. Answer the following questions.

a. Check the HTTP request packet to 93.184.216.34 (ip of [www.example.com](http://www.example.com)). What are the source MAC and destination MAC? You need to check the link layer header in the packet. The source MAC is the MAC of your VM.

b. Does the destination MAC in **a** belong to 93.184.216.34? To find out your answer, run command **arp** to check the arp table of your VM. Is the destination MAC in **a** listed here? If yes, confirm that this MAC does not belong to 93.184.216.34 and instead belong to your router.

c. In the upper protocol field of link layer header of your HTTP request packet, what is the value? What protocol does it represent?